Surya Thirukonda

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Education

Texas A&M University — College Station, TX
Bachelor of Science in Computer Science; GPA: 4.0 — *Expected May 2028*

Experience

Software Engineering Intern, Quantitative Systems — Jun. 2026 – Present
DeKalb Capital Management — *Remote*

- Developed portfolio analytics and stock-screening features for a Dockerized FastAPI dashboard, integrating yfinance and the IBKR API to synchronize positions, calculate time-weighted returns, and serve market data across 100+ tracked equities.
- Improved platform reliability with QuestDB time-series storage, application-level caching, scheduled and on-demand portfolio syncs, API rate limiting, schema validation, and health checks; supported Railway and Cloudflare deployments.

Founding Software Engineer (Part-time) — Feb. 2026 – Present
Revio — *Remote*

- Architected and implemented a Dockerized FastAPI and React/Vite manuscript-processing platform from user flows and interface design through production deployment, launching a live beta with 40+ journal profiles and supporting 200+ end-to-end document conversions.
- Engineered asynchronous parsing, schema validation, and Gotenberg DOCX/PDF transformation pipelines with Supabase persistence and OAuth; deployed containerized services across Railway and Vercel while maintaining 99%+ uptime during the live beta.

Contract Software Engineer — Feb. 2026 – Jun. 2026
Mai Shan Yun — *College Station, TX*

- Won the Mai Shan Yun Challenge at TAMUhack among 40+ teams, converting the prototype into a four-month contract to build and deploy WaiterDock for a local restaurant; implemented Stripe payments, RBAC, order-state APIs, and real-time WebSocket synchronization.
- Built workforce-management features spanning employee authentication, clock-in/clock-out tracking, live task assignment, SMTP alerts, and automated payroll calculations across front-of-house, kitchen, and administrative workflows.

Research & Projects

Undergraduate Researcher (REU) — Jun. 2026 – Present
Portland State University — *Remote*

- Developed leakage-safe walk-forward forecasting across 11 U.S. Treasury maturities and 1-, 5-, and 20-day horizons, benchmarking 7 statistical and machine-learning approaches over 250 rolling forecast origins.
- Implemented HMM/HSMM regime features and Diebold-Mariano evaluation, comparing forecast accuracy across yield-curve regimes and evaluating model performance against statistical and machine-learning baselines.

Software Development Engineer, VEST Project — Jan. 2026 – May 2026
Turtle Robotics — *College Station, TX*

- Engineered REST APIs and cloud pipelines for a canine health-monitoring vest, synchronizing 12 samples per second across 4 biometric streams from an ESP32 through Supabase to a React Native monitoring interface.

RateSignal | *React, Express, PostgreSQL, Prisma, Zod, Redis, Docker, Python* — 2026

- Built and deployed a Treasury forecasting platform serving 1,000+ predictions across 11 maturities, 7 statistical and machine-learning models, and 1-, 5-, and 20-trading-day horizons through authenticated REST APIs and an interactive dashboard.
- Implemented JWT authentication, Prisma schemas, Zod validation, Redis-backed rate limiting, OpenAPI documentation, Docker Compose, and an optimized backend and batch-ingestion pipeline for 600+ predictions per upload, reducing full database seed time from over 50 minutes to approximately 5 seconds.

Data Analyst, Southwest Airlines Partnership — Oct. 2025 – Dec. 2025
Aggie Data Science Club — *College Station, TX*

- Built a SARIMAX baggage-revenue forecasting pipeline using Southwest Airlines-provided data, engineering seasonal, macroeconomic, policy, and operational features to achieve below 12% MAPE on baggage-revenue forecasts across a 24-month horizon.
- Cleaned and aligned 10+ years of monthly revenue and economic data; performed backtesting, residual diagnostics, confidence-interval analysis, and presented revenue-risk scenarios and model findings to Southwest Airlines stakeholders.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, R
Backend/Web: FastAPI, Node.js, Express, React, REST APIs, WebSockets, Prisma, SQLAlchemy, JWT, OAuth
Data/ML: pandas, NumPy, scikit-learn, statsmodels, XGBoost, SARIMAX, VAR, Dynamic Nelson-Siegel, HMM/HSMM
Infrastructure: PostgreSQL, QuestDB, Redis, Supabase, Docker, GitHub Actions, AWS, Railway, Cloudflare, Git, Linux
Certification: AWS Certified Cloud Practitioner